

PET and Gestational Hypertension 02: Placenta, Angiogenic Imbalance, and Maternal Endothelium

Companion reading handout for simultaneous listening.

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Let us start with a patient who looks, at first, like she has a blood pressure problem.

She is thirty weeks. Her BP is one hundred fifty over ninety six. Her ankles are swollen. Her urine protein is rising. The fetus is measuring small. A lab calls with a platelet count that has fallen since last week.

It is tempting to see several separate problems. Hypertension. Proteinuria. Fetal growth restriction. Thrombocytopenia.

But pre-eclampsia is powerful because it turns those separate problems into one story.

The story begins in the placenta, and it declares itself through the maternal endothelium.

In a normal pregnancy, trophoblast cells invade and remodel the spiral arteries. Those spiral arteries are supposed to become low-resistance, high-flow vessels. Gabbe describes the normal placental bed as involving roughly one hundred to one hundred fifty spiral arterioles. The purpose of remodeling is simple: the placenta needs a stable, generous blood supply that is not tightly vasoconstricted like ordinary maternal arterioles.

In early placental disease, that remodeling is incomplete. The deeper myometrial segments of the spiral arteries remain more muscular and more reactive. The placenta receives blood through a higher resistance system. Flow becomes more vulnerable to ischemia, reperfusion, oxidative stress, and malperfusion.

This is why early-onset pre-eclampsia, fetal growth restriction, oligohydramnios, abnormal Dopplers, and abruption often travel together. They are different faces of placental dysfunction.

But placental malperfusion alone is not the whole disease. The placenta then communicates with the mother.

The most important language is angiogenic balance.

Placental growth factor, or PlGF, and vascular endothelial growth factor, or VEGF, help maintain vascular and endothelial health. Soluble fms-like tyrosine kinase one, or sFlt-1, is an antiangiogenic soluble receptor released by the placenta. sFlt-1 binds VEGF and PlGF. When sFlt-1 rises, free VEGF and PlGF fall. The maternal endothelium loses pro-angiogenic support.

That is the biochemical bridge between placenta and mother.

The placenta is not merely failing locally. It is releasing a signal that changes the maternal vascular system.

Endothelial dysfunction then explains the clinical syndrome.

First, vasospasm and increased systemic vascular resistance produce hypertension. The blood pressure is not the disease itself; it is the visible hemodynamic expression of a vascular disorder.

Second, renal endothelial injury produces glomerular endotheliosis. The glomerular filter becomes abnormal, and protein leaks into urine. This is why proteinuria became the classic diagnostic sign. It is a kidney footprint of systemic endothelial disease.

Third, capillary leak produces edema, hemoconcentration, and pulmonary edema risk. This is a crucial bedside point. The patient may look edematous, but she is not a simple fluid-overload patient in the ordinary sense. She can be intravascularly vulnerable and extravascularly overloaded at the same time. If you pour fluid into a leaky endothelial system, some of that fluid goes to the lungs. This is why severe pre-eclampsia is not treated with generous maintenance fluids, and why NICE limits maintenance fluids in severe disease unless there are ongoing losses.

Fourth, platelet activation and microangiopathy produce thrombocytopenia. Platelets are consumed at the injured endothelial surface. When the platelet count falls below one hundred thousand, ACOG and the local Hadassah protocol treat that as a severe feature. But do not wait only for the threshold. A falling trend can tell you that the disease is moving.

Fifth, hepatic endothelial injury and sinusoidal obstruction produce elevated transaminases and right upper quadrant or epigastric pain. That pain is not indigestion until proven otherwise. In the right context, it can represent hepatic capsular stretch, HELLP syndrome, subcapsular hematoma, or impending catastrophe.

Sixth, brain endothelial dysfunction produces headache, visual symptoms, PRES, stroke, and eclampsia. The brain is where the blood pressure threshold becomes especially dangerous. Severe hypertension can overwhelm autoregulation, and the result may be hemorrhage, edema, seizure, or stroke.

Seventh, the fetus shows the placental side of the same disease. Fetal growth restriction, abnormal umbilical artery Doppler, oligohydramnios, placental abruption,

nonreassuring fetal status, and stillbirth are not random complications. They are what happens when placental function cannot keep up.

This is the core map.

Abnormal placentation leads to placental stress. Placental stress leads to antiangiogenic imbalance. Antiangiogenic imbalance injures maternal endothelium. Endothelial dysfunction creates hypertension, proteinuria, platelet consumption, liver injury, pulmonary edema, neurologic disease, and fetal placental disease.

Refining the Model

Now let us make the model more sophisticated.

Not all pre-eclampsia is the same.

Early-onset pre-eclampsia, especially before thirty four weeks, is more often dominated by placental maldevelopment and placental insufficiency. It is more strongly associated with fetal growth restriction and abnormal biomarkers. Late-onset or term pre-eclampsia may have more to do with the interaction between the normal placenta and a maternal cardiovascular system that cannot meet the demands of late pregnancy. It is still endothelial disease, but the upstream balance may be different.

This explains why prediction is better for preterm pre-eclampsia than for term pre-eclampsia. It also explains why aspirin mainly prevents preterm disease. Aspirin is most useful when started early enough to influence placentation and platelet-inflammatory pathways. It is not magic. It cannot prevent every late maternal cardiovascular mismatch.

Angiogenic Testing

The angiogenic tests fit naturally into this model.

The sFlt-1 to PlGF ratio is not a proteinuria test. It is not asking, is the kidney leaking protein? It is asking, does the maternal circulation show the antiangiogenic pattern of placental pre-eclampsia biology?

That is why the PROGNOSIS trial matters. In women with suspected pre-eclampsia between twenty four and thirty six plus six weeks, an sFlt-1 to PlGF ratio of thirty eight or lower had a very high negative predictive value for ruling out pre-eclampsia in the short term. The ratio was best as reassurance that imminent pre-eclampsia was unlikely. A high ratio was concerning, but much less perfect as a rule-in test.

So the biomarker teaching line is this: low ratio, strong short-term reassurance. High ratio, increased concern, not a diagnosis by itself.

This also explains why biomarkers cannot override the bedside. If a patient has pulmonary edema, platelets of seventy thousand, rising creatinine, severe headache, or eclampsia, she has dangerous organ disease. A reassuring biomarker does not make her safe.

Closing Synthesis

Let us close by making the pathophysiology memorable.

Pre-eclampsia is not simply high blood pressure in pregnancy. It is a placental-maternal endothelial syndrome.

The placenta starts the conversation. The endothelium spreads it through the body. The organs tell you where the injury is landing.

Resident Reflex

The resident reflex is this: every sign in pre-eclampsia should be translated back to endothelial disease. Proteinuria is renal endothelial injury. Low platelets are microangiopathy. Pulmonary edema is capillary leak. Headache and visual symptoms are cerebral endothelial danger. Fetal growth restriction is placental failure. When you understand that, the syndrome stops being a list and becomes a map.